JAMA Internal Medicine | Original Investigation | CLIMATE CHANGE AND HEALTH

Comparative Effectiveness and Safety of LAMA-LABA Inhalers in Chronic Obstructive Pulmonary Disease

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IMPORTANCE Dual bronchodilator therapy with a long-acting muscarinic antagonist (LAMA) and a long-acting β_2-agonist (LABA) is recommended for most patients with symptomatic chronic obstructive pulmonary disease (COPD). Fixed-dose LAMA-LABA therapies are available in metered-dose, dry powder, and soft mist inhalers. However, metered-dosed inhalers are associated with greater greenhouse gas emissions than either dry powder or soft mist inhalers, and questions persist about potential intraclass differences among LAMA-LABAs given variability in their active ingredients, dosing schedules, and delivery devices.

OBJECTIVE To evaluate the comparative effectiveness and safety of once-daily umeclidinium-vilanterol dry powder inhalers, twice-daily glycopyrrolate-formoterol metered-dosed inhalers, and once-daily tiotropium-olodaterol soft mist inhalers.

DESIGN, SETTING, AND PARTICIPANTS This observational active-comparator study analyzed claims of patients (≥40 years) newly treated with LAMA-LABA inhalers and continuously enrolled in a large commercial health insurance or Medicare Advantage plan during the 183-day baseline period. Patients were propensity score matched 1:1 into 3 cohorts with index dates ranging from May 1, 2016, to February 28, 2025. Data were analyzed from July to August 2025.

EXPOSURES Patients treated with umeclidinium-vilanterol, glycopyrrolate-formoterol, or tiotropium-olodaterol fixed-dose inhalers.

MAIN OUTCOMES AND MEASURES Time to the first moderate or severe COPD exacerbation, major adverse cardiovascular event, urinary tract infection, and pneumonia hospitalization.

RESULTS The cohorts included 9479 matched pairs of patients receiving umeclidinium-vilanterol vs glycopyrrolate-formoterol (mean age, 68.9 [SD, 9.0] years; 10 319 women [54.4%]; 8636 men [45.6%]), 9598 receiving tiotropium-olodaterol vs glycopyrrolate-formoterol (mean age, 69.2 [SD, 8.7] years; 10 513 women [54.8%]; 8680 men [45.2%]), and 36 740 receiving umeclidinium-vilanterol vs tiotropium-olodaterol (mean age, 71.5 [SD, 8.4] years; 39 429 women [53.7%]; 34 044 men [46.3%]). Umeclidinium-vilanterol was associated with a 14% lower hazard of a first moderate or severe COPD exacerbation than glycopyrrolate-formoterol (hazard ratio [HR], 0.86; 95% CI, 0.81-0.91; number needed to treat [NNT], 17) and was associated with a 3% lower hazard than tiotropium-olodaterol (HR, 0.97; 95% CI, 0.94-0.99; NNT, 100). Tiotropium-olodaterol was associated with a 6% lower hazard of a first moderate or severe COPD exacerbation than glycopyrrolate-formoterol (HR, 0.94; 95% CI, 0.89-1.00). Similar risks of first major adverse cardiovascular event, urinary tract infection, and pneumonia hospitalization were observed among patients in all 3 cohorts.

CONCLUSIONS AND RELEVANCE This cohort study found that umeclidinium-vilanterol was associated with improved clinical outcomes compared with glycopyrrolate-formoterol and tiotropium-olodaterol. Patients, prescribers, and health systems may consider once-daily umeclidinium-vilanterol dry powder inhalers over alternatives among new users of LAMA-LABA therapy.

JAMA Intern Med. doi:10.1001/jamainternmed.2025.8087
Published online February 23, 2026.

Supplemental content

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Dual bronchodilators with a long-acting muscarinic antagonist (LAMA) and a long-acting β_2-agonist (LABA) are widely used in the management of chronic obstructive pulmonary disease (COPD).[1,2] The US Food and Drug Administration has approved several fixed-dose LAMA-LABA inhalers over the past decade,[3,4] each demonstrating improvements in lung function, symptoms, and acute exacerbations compared with placebo or single-agent therapy.[5-14] Several randomized clinical trials and longitudinal database studies have also found LAMA-LABA therapy to be associated with fewer COPD exacerbations and episodes of pneumonia compared with inhaled corticosteroid (ICS)-LABA therapy.[15-19] Consensus international guidelines now recommend LAMA-LABA inhalers as a first-line treatment for most patients with COPD experiencing high symptom burden.[1]

LAMA-LABA therapies are available on the US market as metered-dose inhalers, dry powder inhalers, and soft mist inhalers. Head-to-head comparisons of these products in randomized trials using surrogate end points,[20-23] indirect meta-analyses,[24-26] and longitudinal database studies from the United Kingdom[27] and Taiwan[28] have generated mixed findings about potential intraclass differences. Furthermore, prior studies have been limited by low power, exclusion of certain device types, and/or questions of generalizability to US patients. Current COPD guidelines express no preference for one LAMA-LABA inhaler over another despite having different active ingredients, dosing schedules, and delivery devices.[1]

Health systems have sought to promote use of dry powder and soft mist inhalers over metered-dosed inhalers[29-32] because metered-dosed inhalers rely on hydrofluorocarbon propellants, which are potent greenhouse gases.[33] Dry powder and soft mist inhalers, by contrast, are propellant-free and associated with just one-twentieth the emissions of metered-dosed inhalers.[34] Despite the push for wider adoption of dry powder and soft mist inhalers to mitigate the effects of climate change,[35-37] scant large-scale data are available directly comparing LAMA-LABA inhalers to one another.

To address the current gap in evidence, this real-world, comparative effectiveness and safety study analyzes the 3 most widely prescribed LAMA-LABA regimens in the US: once-daily umeclidinium-vilanterol dry powder inhalers (Anoro Ellipta), twice-daily glycopyrrolate-formoterol metered-dosed inhalers (Bevespi Aerosphere), and once-daily tiotropium-olodaterol soft mist inhalers (Stiolto Respimat). Findings from this study may support more informed COPD treatment decisions, with implications for both clinical management and environmental stewardship.

Key Points

Question Are different fixed-dose long-acting muscarinic antagonist (LAMA)–long-acting β_2-agonists (LABA) associated with differential clinical outcomes among patients with chronic obstructive pulmonary disease (COPD) treated in routine clinical practice?

Findings In this active comparator new-user cohort study of 55 817 matched pairs, once-daily umeclidinium-vilanterol dry powder inhalers were associated with a 14% lower hazard of moderate or severe COPD exacerbations than twice-daily glycopyrrolate-formoterol metered-dose inhalers and a 3% lower hazard than once-daily tiotropium-olodaterol soft mist inhalers.

Meaning Patients, prescribers, and health systems may consider once-daily umeclidinium-vilanterol dry powder inhalers among new users of LAMA-LABA therapy.

Methods

Study Population

We used data from Optum's deidentified Clinformatics Data Mart Database, which includes demographic information, inpatient and outpatient medical claims, pharmacy claims, and laboratory test results for more than 95 million individuals in the US with commercial and Medicare Advantage health plans. We constructed 3 separate cohorts for analysis: (1) patients initiating treatment with umeclidinium-vilanterol (exposure) vs glycopyrrolate-formoterol (referent) between January 1, 2017, and February 28, 2025; (2) patients initiating treatment with tiotropium-olodaterol (exposure) vs glycopyrrolate-formoterol (referent) between January 1, 2017, and February 28, 2025; and (3) patients initiating treatment with umeclidinium-vilanterol (exposure) vs tiotropium-olodaterol (referent) between May 1, 2016, and February 28, 2025 (eMethods in Supplement 1).

All treatment episodes with the medications of interest were identified. Eligible patients were required to be at least 40 years of age (analogous to prior trials[20-23]), have continuous health plan enrollment during the 183 days prior to initiation, and have an active COPD diagnosis, which was identified by having 3 outpatient claims or 1 inpatient claim for COPD in the 3 years before and including the date of treatment initiation (positive predictive value [PPV], 0.83; 95% CI, 0.72-0.89).[38] COPD was defined by *International Classification of Diseases, Ninth Revision, Clinical Modification* codes 491.xx, 492.xx, 493.2x, or 496 and *International Statistical Classification of Diseases, Tenth Revision, Clinical Modification (ICD-10-CM)* codes J41.x, J42, J43.x, or J44.x.

The study was approved by the Mass General Brigham Institutional Review Board (2023P000164) and was preregistered in the Open Science Framework Real-World Evidence Registry.[39] Informed consent was not required because data were deidentified. This study adheres to the Reporting of Studies Conducted Using Observational Routinely Collected Health Data (RECORD) guidelines.

To identify new use of LAMA-LABA inhalers, we excluded patients who had filled any LAMA-LABA inhaler in the 183 days prior or had received dual LAMA-LABA therapy via 2 separate inhalers filled within 30 days of one another. Patients were permitted to receive other maintenance inhalers during the baseline period. However, we excluded episodes in which a prescription for an ICS or any combination of ICS, LAMA, or LABA agents was filled on the date of treatment initiation (other than the index medications under assessment) to ensure that we were analyzing only episodes of new LAMA-LABA therapy. A detailed list of claims codes used to

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define inclusion and exclusion criteria is provided in eTable 1 in Supplement 1. The first treatment episode for each patient that met all eligibility criteria was used for analysis; patients could only enter each of the 3 cohorts once.

Covariate Assessment

Potential confounders were assessed during the 183-day baseline period prior to and including patients' cohort entry dates. Patient age, sex, US region of residence, year of cohort entry, and season of cohort entry were measured on the index date. Clinical characteristics, including lung disease related to COPD, comorbidities, health care utilization, and receipt of other medications, were evaluated during the baseline period. History of prior asthma diagnosis was assessed across the entirety of each patient's available claims data, while eosinophil counts were based on the most recent recorded value within the baseline period. All covariates were included in the propensity score model and are defined in eTable 2 in Supplement 1. The timing of variable assessments is illustrated in eFigure 1 in Supplement 1; eFigure 2 in Supplement 1 provides a directed acyclic graph, including potential unmeasured factors.

Outcome Definitions and Follow-Up

Follow-up for outcomes began one day after the cohort entry date and continued for up to a year. The primary effectiveness outcome was time to first moderate or severe COPD exacerbation. Moderate exacerbations were defined as a prescription fill for an oral glucocorticoid (ie, prednisone, prednisolone, methylprednisolone) with a supply of 5 to 14 days (PPV, 0.73).[40] Severe exacerbations were defined as hospitalizations with a primary diagnosis of COPD using *ICD-10-CM* codes listed above (PPV, 0.86).[41] Exacerbations were considered distinct if separated by at least 14 days—events that met the criteria for both moderate and severe exacerbation within this period were counted as 1 severe exacerbation on the earliest event date. In secondary analyses, we separately analyzed time to first moderate and first severe COPD exacerbation.

Safety outcomes included time to first major adverse cardiovascular event (MACE) (defined as all-cause mortality, acute myocardial infarction, or stroke), first outpatient urinary tract infection (UTI), and first pneumonia-related hospitalization. These adverse effects were selected because the first 2 are known risks of LAMA-LABAs (adrenergic agents may predispose patients to cardiac events and anticholinergic medications to UTI), whereas low rates of pneumonia are a particular advantage of the class compared with ICS-containing regimens. See eTable 3 in Supplement 1 for claims-based definitions of outcomes and performance characteristics.

We conducted as-treated analyses. Accordingly, patients were censored at the earliest occurrence of the outcome of interest, death, treatment discontinuation (defined as a gap >60 days between prescription fills, with a 60-day exposure risk window), end of data availability, end of the study period, switch to another LAMA-LABA inhaler, or initiation of another maintenance inhaler (eg, ICS, ICS-LAMA-LABA). Patients who were not censored on the cohort entry date were included in analyses.

Statistical Analysis

We conducted 1:1 propensity score matching using nearest-neighbor matching without replacement and a caliper of 1%. Propensity scores were estimated using logistic regression, and the model included all covariates specified in the eMethods section in Supplement 1. For covariates with missing data, a missing indicator was used. In the matched cohorts, hazard ratios (HRs) and 95% CIs were estimated using Cox proportional hazards models. For estimates where 95% CIs excluded the null value, we calculated the number needed to treat (NNT) as the inverse of the absolute risk difference at 365 days.

In a secondary analysis, we compared the annual rates of total moderate or severe COPD exacerbations in the 365 days of follow-up using a negative binomial regression with an offset for the log of observed person-time.

Subgroup and Sensitivity Analyses

We evaluated prespecified subgroups defined based on indicators of disease severity (eg, occurrence of moderate or severe exacerbations during the baseline period), comorbid asthma, and receipt of spirometry and whether the index prescription was written by a pulmonologist. Subgroup characteristics were assessed during the baseline period, except for asthma diagnosis, which was evaluated using patients' full available claims history. Analyses were conducted separately for those with and without each subgroup characteristic. Propensity scores were reestimated and patients were rematched between treatment groups within each subgroup.

We also conducted several sensitivity analyses to test the robustness of our findings. First, we varied the allowable grace period between prescription fills used to define treatment discontinuation from 60 days (primary analysis) to 30 and 90 days. Second, we conducted an as-started analysis, in which follow-up continued for 1 year regardless of treatment discontinuation or switching. Third, we restricted follow-up to the first 90 days after cohort entry. Fourth, we trimmed the top and bottom 2.5% of propensity scores in the unmatched cohort and adjusted outcome models for the decile of patients' propensity score to analyze the data with increased sample size compared with the primary analysis. Fifth, we evaluated the impact of alternative outcome definitions for moderate and severe COPD exacerbations. Sensitivity definitions for a moderate exacerbation included (1) requiring an outpatient or emergency department visit prior to an oral glucocorticoid fill; (2) requiring both an oral glucocorticoid and an antibiotic fill; and (3) permitting either an oral glucocorticoid or a respiratory antibiotic fill (eTable 4 in Supplement 1). For severe exacerbations, we tested a narrower definition limited to hospitalizations with *ICD-10-CM* codes (J44.0, J44.1) for an acute COPD exacerbation in any diagnostic position (eMethods in Supplement 1). Finally, as an additional prespecified sensitivity analysis, we conducted 1:1 high-dimensional propensity score matching using an automated algorithm to empirically select covariates from diagnosis codes, procedures, and pharmacy claims for inclusion in the propensity score model. The Bross formula was used for covariate ranking, and the number of covariates was automatically chosen by the algorithm.[42-44]

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We used a Cox proportional hazards model to estimate HRs and 95% CIs.

All analyses were performed separately within each cohort using the Aetion Substantiate (Datavant) with R version 4.73 integration and in RStudio (R Statistical Software, version 4.4.1; eMethods in Supplement 1). Data were analyzed from July to August 2025.

Results

Study Population

Of the 55 817 1:1 propensity score matched pairs, 9479 were in the umeclidinium-vilanterol vs glycopyrrolate-formoterol cohort (mean age, 68.9 [SD, 9.0] years; 10 319 women [54.4%]; 8636 men [45.6%]); 9598 in the tiotropium-olodaterol vs glycopyrrolate-formoterol cohort (mean age, 69.2 [SD, 8.7] years; 10 513 women [54.8%]; 8680 men [45.2%]); and 36 740 in the umeclidinium-vilanterol vs tiotropium-olodaterol cohort (mean age, 71.5 [SD, 8.4] years; 39 429 women [53.7%]; 34 044 men [46.3%]); **Table 1** and eFigure 3 in Supplement 1). Prior to matching, the C statistics of the propensity score model were 0.64 in the umeclidinium-vilanterol vs glycopyrrolate-formoterol cohort, 0.70 in the tiotropium-olodaterol vs glycopyrrolate-formoterol cohort, and 0.62 in the umeclidinium-vilanterol vs tiotropium-olodaterol cohort, and the mean absolute standardized differences among baseline characteristics were 0.05, 0.04, and 0.03, respectively (eTable 5 in Supplement 1). After matching, the respective C statistics were 0.53, 0.53, and 0.51, and mean absolute standardized differences were less than 0.01 in all cohorts. The distributions of propensity scores before and after matching are presented in eFigure 4 in Supplement 1.

Primary Effectiveness Outcomes

The incidence rate of first moderate or severe COPD exacerbation was 0.75 per person-year in the matched cohort comparing umeclidinium-vilanterol with glycopyrrolate-formoterol, 0.78 per person-year in the matched cohort comparing tiotropium-olodaterol with glycopyrrolate-formoterol, and 0.68 per person-year in the matched cohort comparing umeclidinium-vilanterol with tiotropium-olodaterol (**Table 2**). Median follow-up times and reasons for discontinuation are given in eTable 6 in Supplement 1.

Umeclidinium-vilanterol was associated with a 14% lower hazard of a first moderate or severe COPD exacerbation compared with glycopyrrolate-formoterol (HR, 0.86; 95% CI, 0.81-0.91; NNT, 17) and a 3% lower hazard compared with tiotropium-olodaterol (HR, 0.97; 95% CI, 0.94-0.99; NNT, 100). Tiotropium-olodaterol was associated with a 6% lower hazard of first moderate or severe COPD exacerbation compared with glycopyrrolate-formoterol (HR, 0.94; 95% CI, 0.89-1.00). Kaplan-Meier survival plots are presented in eFigure 5 in Supplement 1. Sensitivity analyses yielded similar findings to the primary analyses (**Figure 1**).

Secondary Effectiveness Outcomes

Initiation of umeclidinium-vilanterol was associated with a 14% lower hazard of first moderate COPD exacerbation compared with initiation of glycopyrrolate-formoterol (HR, 0.86; 95% CI, 0.81-0.91; NNT, 17; Table 2). No differences were observed in the hazards of first moderate exacerbation when comparing tiotropium-olodaterol to glycopyrrolate-formoterol or umeclidinium-vilanterol to glycopyrrolate-formoterol. There were also no differences in the risk of first severe COPD exacerbation between any of the inhalers. Sensitivity analyses yielded similar results (eFigure 6 and eFigure 7 in Supplement 1).

Annual Rate of Exacerbations

Umeclidinium-vilanterol was associated with a 14% reduction in the annual rate of moderate or severe COPD exacerbations over 1 year of follow-up compared with glycopyrrolate-formoterol (incidence rate ratio [RR], 0.86; 95% CI, 0.81-0.91) and an 8% reduction compared with tiotropium-olodaterol (incidence RR, 0.92; 95% CI, 0.89-0.95). There was a 6% reduction in the annual rate of exacerbations among patients receiving tiotropium-olodaterol compared with glycopyrrolate-formoterol (incidence RR, 0.94; 95% CI, 0.89-1.00).

Subgroup Analyses

Umeclidinium-vilanterol was associated with a reduced hazard of first moderate or severe COPD exacerbation compared with glycopyrrolate-formoterol across nearly all subgroups (**Figure 2**). Effect sizes were smaller in the cohorts comparing tiotropium-olodaterol vs glycopyrrolate-formoterol and umeclidinium-vilanterol vs tiotropium-olodaterol; point estimates across all subgroups suggested benefit from the comparator inhaler, although 95% CIs in several subgroups included the null value.

Safety Outcomes

The incidence rates of safety outcomes were similar for each matched cohort: 0.10 per person-year MACE, 0.02 per person-year UTI, and 0.10 per person-year pneumonia hospitalization (Table 2). In propensity score-matched analyses, similar risks of all 3 adverse effects were observed among the comparator and referent groups. Sensitivity analyses showed similar results to the primary estimates and are presented in eFigure 8 (MACE), eFigure 9 (UTI), and eFigure 10 in Supplement 1 (pneumonia hospitalization).

Discussion

In this observational comparative effectiveness and safety study, initiation of once-daily umeclidinium-vilanterol dry powder inhalers was associated with a slightly reduced risk of first moderate or severe COPD exacerbation compared with initiation of twice-daily glycopyrrolate-formoterol metered-dosed inhalers and once-daily tiotropium-olodaterol soft mist inhalers. Similar risks were observed for patients receiving tiotropium-olodaterol and glycopyrrolate-formoterol, although point estimates favored the once-daily soft mist inhalers over the twice daily metered-dosed inhalers. These findings were robust across sensitivity and subgroup analyses in each cohort. No differences in the safety outcomes were observed when comparing any of the 3 therapies.

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Table 1. Baseline Characteristics of Patients Initiating Umeclidinium-Vilanterol, Glycopyrrolate-Formoterol, or Tiotropium-Olodaterol in Propensity Score Matched Cohorts

Characteristic, No. (%)[a]	Cohort 1			Cohort 2			Cohort 3		
	Umeclidinium-vilanterol vs glycopyrrolate-formoterol (2017-2025; n = 9479 matched pairs)[b]			Tiotropium-olodaterol vs glycopyrrolate-formoterol (2017-2025; n = 9598 matched pairs)[b]			Umeclidinium-vilanterol vs tiotropium-olodaterol (2016-2025; n = 36 740 matched pairs)[b]		
	Umeclidinium-vilanterol	Glycopyrrolate-formoterol	Absolute standardized difference	Tiotropium-olodaterol	Glycopyrrolate-formoterol	Absolute standardized difference	Umeclidinium-vilanterol	Tiotropium--olodaterol	Absolute standardized difference
Age, mean (SD), y	68.98 (9.07)	68.90 (8.96)	0.009	69.25 (8.68)	69.15 (8.80)	0.011	71.54 (8.51)	71.45 (8.27)	0.011
Female	5150 (54.3)	5169 (54.5)	0.004	5276 (55.0)	5237 (54.6)	0.008	19 723 (53.7)	19 706 (53.6)	0.002
Male	4328 (45.7)	4308 (45.4)	0.006	4321 (45.0)	4359 (45.4)	0.008	17 013 (46.3)	17 031 (46.4)	0.002
Region									
Northeast	937 (9.9)	987 (10.4)	0.017	1073 (11.2)	1023 (10.7)	0.016	3811 (10.4)	3815 (10.4)	<0.001
Midwest	1891 (19.9)	1841 (19.4)	0.013	1853 (19.3)	1862 (19.4)	0.003	7127 (19.4)	7232 (19.7)	0.008
South	5054 (53.3)	5019 (52.9)	0.008	5019 (52.3)	5053 (52.6)	0.006	17 097 (46.5)	17 118 (46.6)	0.002
West	1596 (16.8)	1631 (17.2)	0.011	1651 (17.2)	1659 (17.3)	0.003	8683 (23.6)	8559 (23.3)	0.007
Year of cohort entry date									
2016							1756 (4.8)	1721 (4.7)	0.005
2017	1367 (14.4)	1433 (15.1)	0.020	1371 (14.3)	1432 (14.9)	0.017	3933 (10.7)	3875 (10.5)	0.006
2018	2208 (23.3)	2149 (22.7)	0.014	2105 (21.9)	2064 (21.5)	0.010	3712 (10.1)	3732 (10.2)	0.003
2019	1810 (19.1)	1781 (18.8)	0.008	1819 (19.0)	1816 (18.9)	0.003	4200 (11.4)	4131 (11.2)	0.006
2020	1169 (12.3)	1165 (12.3)	<0.001	1224 (12.8)	1215 (12.7)	0.003	3573 (9.7)	3540 (9.6)	0.003
2021	845 (8.9)	858 (9.1)	0.007	885 (9.2)	906 (9.4)	0.007	3751 (10.2)	3795 (10.3)	0.003
2022	617 (6.5)	652 (6.9)	0.016	645 (6.7)	666 (6.9)	0.008	4381 (11.9)	4439 (12.1)	0.006
2023	721 (7.6)	716 (7.6)	<0.001	762 (7.9)	739 (7.7)	0.007	5561 (15.1)	5623 (15.3)	0.006
2024-2025[c]	742 (7.8)	725 (7.7)	0.004	787 (8.2)	760 (7.9)	0.011	5873 (16.0)	5884 (16.0)	<0.001
Seasons of initiation									
Winter	2205 (23.3)	2181 (23.0)	0.007	2267 (23.6)	2211 (23.0)	0.014	8738 (23.8)	8786 (23.9)	0.002
Spring	2198 (23.2)	2270 (23.9)	0.016	2249 (23.4)	2318 (24.2)	0.019	9056 (24.6)	9054 (24.6)	<0.001
Summer	2490 (26.3)	2479 (26.2)	0.002	2466 (25.7)	2486 (25.9)	0.005	9183 (25.0)	9136 (24.9)	0.002
Fall	2586 (27.3)	2549 (26.9)	0.009	2616 (27.3)	2583 (26.9)	0.009	9763 (26.6)	9764 (26.6)	<0.001
Baseline lung disease, COPD exacerbations									
Moderate exacerbations									
0	6372 (67.2)	6395 (67.5)	0.006	6515 (67.9)	6518 (67.9)	<0.001	25 561 (69.6)	25 571 (69.6)	<0.001
1	2202 (23.2)	2184 (23.0)	0.005	2147 (22.4)	2182 (22.7)	0.007	8062 (21.9)	8069 (22.0)	0.002
≥2	905 (9.5)	900 (9.5)	<0.001	936 (9.8)	898 (9.4)	0.014	3117 (8.5)	3100 (8.4)	0.004
Severe exacerbations									
0	9008 (95.0)	9037 (95.3)	0.014	9152 (95.4)	9167 (95.5)	0.005	35 131 (95.6)	35 133 (95.6)	<0.001
1	436 (4.6)	414 (4.4)	0.010	421 (4.4)	405 (4.2)	0.010	1491 (4.1)	1475 (4.0)	0.005
≥2	35 (0.4)	28 (0.3)	0.017	25 (0.3)	26 (0.3)	<0.001	118 (0.3)	132 (0.4)	0.017

(continued)

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Table 1. Baseline Characteristics of Patients Initiating Umeclidinium-Vilanterol, Glycopyrrolate-Formoterol, or Tiotropium-Olodaterol in Propensity Score Matched Cohorts (continued)

Characteristic, No. (%)[a]	Cohort 1 Umeclidinium-vilanterol vs glycopyrrolate-formoterol (2017-2025; n = 9479 matched pairs)[b]			Cohort 2 Tiotropium-olodaterol vs glycopyrrolate-formoterol (2017-2025; n = 9598 matched pairs)[b]			Cohort 3 Umeclidinium-vilanterol vs tiotropium-olodaterol (2016-2025; n = 36 740 matched pairs)[b]		
	Umeclidinium-vilanterol	Glycopyrrolate-formoterol	Absolute standardized difference	Tiotropium-olodaterol	Glycopyrrolate-formoterol	Absolute standardized difference	Umeclidinium-vilanterol	Tiotropium--olodaterol	Absolute standardized difference
Moderate or severe exacerbations									
0	6086 (64.2)	6125 (64.6)	0.008	6236 (65.0)	6254 (65.2)	0.004	24 513 (66.7)	24 521 (66.7)	<0.001
1	2347 (24.7)	2324 (24.5)	0.005	2295 (23.9)	2321 (24.2)	0.007	8668 (23.6)	8658 (23.6)	<0.001
≥2	1046 (11.0)	1030 (10.9)	0.003	1067 (11.1)	1023 (10.7)	0.013	3559 (9.7)	3561 (9.7)	<0.001
SAMAs									
0	9246 (97.5)	9252 (97.6)	0.006	9375 (97.7)	9370 (97.6)	0.007	35 952 (97.9)	35 953 (97.9)	<0.001
1	137 (1.4)	135 (1.4)	<0.001	129 (1.3)	133 (1.4)	0.009	423 (1.2)	418 (1.1)	0.009
≥2	96 (1.0)	92 (1.0)	<0.001	94 (1.0)	95 (1.0)	<0.001	365 (1.0)	369 (1.0)	<0.001
SABAs									
0	4055 (42.8)	4050 (42.7)	0.002	4063 (42.3)	4145 (43.2)	0. 018	15 857 (43.2)	15 868 (43.2)	<0.001
1	2415 (25.5)	2401 (25.3)	0.005	2477 (25.8)	2422 (25.2)	0.014	9895 (26.9)	9875 (26.9)	<0.001
≥2	3009 (31.7)	3028 (31.9)	0.004	3058 (31.9)	3031 (31.6)	0.006	10 988 (29.9)	10 997 (29.9)	<0.001
SAMA-SABAs									
0	7986 (84.2)	8000 (84.4)	0.005	8068 (84.1)	8119 (84.6)	0.014	31 192 (84.9)	31 197 (84.9)	<0.001
1	764 (8.1)	731 (7.7)	0.015	765 (8.0)	731 (7.6)	0.015	2790 (7.6)	2791 (7.6)	<0.001
≥2	729 (7.7)	748 (7.9)	0.007	765 (8.0)	748 (7.8)	0.007	2758 (7.5)	2752 (7.5)	<0.001
COPD antibiotic fills									
0	4656 (49.1)	4675 (49.3)	0.004	4721 (49.2)	4786 (49.9)	0.014	19 379 (52.7)	19 506 (53.1)	0.008
1	2483 (26.2)	2448 (25.8)	0.009	2526 (26.3)	2474 (25.8)	0.011	9037 (24.6)	8995 (24.5)	0.002
2	1240 (13.1)	1242 (13.1)	<0.001	1230 (12.8)	1239 (12.9)	0.003	4487 (12.2)	4410 (12.0)	0.006
≥3	1100 (11.6)	1114 (11.8)	0.006	1121 (11.7)	1099 (11.5)	0.006	3837 (10.4)	3829 (10.4)	<0.001
Pneumonia hospitalization	657 (6.9)	607 (6.4)	0.020	593 (6.2)	607 (6.3)	0.004	2778 (7.6)	2772 (7.5)	0.004
Asthma[d]	3163 (33.4)	3167 (33.4)	<0.001	3251 (33.9)	3175 (33.1)	0.017	10 606 (28.9)	10 528 (28.7)	0.004
CPAP or BiPAP	894 (9.4)	879 (9.3)	0.003	885 (9.2)	896 (9.3)	0.003	3532 (9.6)	3460 (9.4)	0.007
Spirometry	3468 (36.6)	3440 (36.3)	0.006	3384 (35.3)	3464 (36.1)	0.017	13 928 (37.9)	13 930 (37.9)	<0.001
Pulmonologist prescribed	1241 (13.1)	1266 (13.4)	0.009	1348 (14.0)	1302 (13.6)	0.012	5533 (15.1)	5577 (15.2)	0.003
Smoking history	8013 (84.5)	7979 (84.2)	0.008	8098 (84.4)	8113 (84.5)	0.003	31 444 (85.6)	31 435 (85.6)	<0.001
Pulmonary rehabilitation	81 (0.9)	77 (0.8)	0.011	88 (0.9)	84 (0.9)	<0.001	311 (0.8)	311 (0.8)	<0.001
LABA	93 (1.0)	101 (1.1)	0.010	89 (0.9)	100 (1.0)	0.010	397 (1.1)	395 (1.1)	<0.001
LAMA	1255 (13.2)	1252 (13.2)	<0.001	1239 (12.9)	1254 (13.1)	0.006	7187 (19.6)	7209 (19.6)	<0.001
ICS	602 (6.3)	607 (6.4)	0.004	585 (6.1)	602 (6.3)	0.008	2021 (5.5)	1993 (5.4)	0.004
ICS-LABA	1983 (20.9)	1995 (21.0)	0.002	2012 (21.0)	1953 (20.3)	0.017	6233 (17.0)	6167 (16.8)	0.005

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Table 1. Baseline Characteristics of Patients Initiating Umeclidinium-Vilanterol, Glycopyrrolate-Formoterol, or Tiotropium-Olodaterol in Propensity Score Matched Cohorts (continued)

Characteristic, No. (%)[a]	Cohort 1 Umeclidinium-vilanterol vs glycopyrrolate-formoterol (2017-2025; n = 9479 matched pairs)[b]			Cohort 2 Tiotropium-olodaterol vs glycopyrrolate-formoterol (2017-2025; n = 9598 matched pairs)[b]			Cohort 3 Umeclidinium-vilanterol vs tiotropium-olodaterol (2016-2025; n = 36 740 matched pairs)[b]		
	Umeclidinium-vilanterol	Glycopyrrolate-formoterol	Absolute standardized difference	Tiotropium-olodaterol	Glycopyrrolate-formoterol	Absolute standardized difference	Umeclidinium-vilanterol	Tiotropium--olodaterol	Absolute standardized difference
ICS-LAMA-LABA	567 (6.0)	525 (5.5)	0.021	526 (5.5)	517 (5.4)	0.004	2087 (5.7)	2042 (5.6)	0.004
Oral steroid (>30-d supply)	574 (6.1)	586 (6.2)	0.004	622 (6.5)	582 (6.1)	0.016	2034 (5.5)	2052 (5.6)	0.004
Azithromycin (>30-d supply)	119 (1.3)	133 (1.4)	0.009	140 (1.5)	130 (1.4)	0.008	356 (1.0)	353 (1.0)	<0.001
Roflumilast (>30-d supply)	89 (0.9)	96 (1.0)	0.010	104 (1.1)	97 (1.0)	0.010	265 (0.7)	258 (0.7)	<0.001
Events in 30 d prior to entry									
Moderate or severe COPD exacerbation	645 (6.8)	651 (6.9)	0.004	662 (6.9)	658 (6.9)	<0.001	2557 (7.0)	2540 (6.9)	0.004
COPD antibiotic fill	1874 (19.8)	1861 (19.6)	0.005	1821 (19.0)	1832 (19.1)	0.003	6462 (17.6)	6395 (17.4)	0.005
Baseline eosinophils									
CBC with differential	5921 (62.5)	5897 (62.2)	0.006	5971 (62.2)	5976 (62.3)	0.002	23 790 (64.8)	23 727 (64.6)	<0.001
Eosinophil categories									
≤100 cell/μL	882 (9.3)	871 (9.2)	0.003	876 (9.1)	881 (9.2)	0.003	3346 (9.1)	3297 (9.0)	0.003
>100-≤ 300 cell/μL	1287 (13.6)	1247 (13.2)	0.012	1237 (12.9)	1257 (13.1)	0.006	4900 (13.3)	4855 (13.2)	0.003
>300 cell/μL	446 (4.7)	443 (4.7)	<0.001	434 (4.5)	450 (4.7)	0.010	1767 (4.8)	1799 (4.9)	0.005
Missing	6864 (72.4)	6918 (73.0)	0.013	7051 (73.5)	7010 (73.0)	0.011	26 727 (72.7)	26 789 (72.9)	0.004
Other comorbidities									
Comorbidity score, mean (SD)	3.15 (2.87)	3.12 (2.84)	0.011	3.14 (2.80)	3.13 (2.84)	0.004	3.43 (3.00)	3.41 (2.98)	0.007
Frailty score, mean (SD)	0.19 (0.06)	0.19 (0.06)	<0.001	0.19 (0.06)	0.19 (0.06)	<0.001	0.19 (0.06)	0.19 (0.06)	<0.001
Obstructive sleep apnea	1931 (20.4)	1888 (19.9)	0.012	1923 (20.0)	1923 (20.0)	<0.001	7690 (20.9)	7632 (20.8)	0.002
Hypertension	6972 (73.6)	6902 (72.8)	0.018	7013 (73.1)	6995 (72.9)	0.005	27 250 (74.2)	27 215 (74.1)	0.002
Diabetes	2900 (30.6)	2865 (30.2)	0.009	2892 (30.1)	2885 (30.1)	<0.001	10 864 (29.6)	10 863 (29.6)	<0.001
Obesity	1851 (19.5)	1829 (19.3)	0.005	1884 (19.6)	1856 (19.3)	0.008	7148 (19.5)	7120 (19.4)	0.003
CAD, CHD, or IHD	2974 (31.4)	2954 (31.2)	0.004	3019 (31.5)	3018 (31.4)	0.002	12 492 (34.0)	12 415 (33.8)	0.004
PVD	2117 (22.3)	2109 (22.2)	0.002	2179 (22.7)	2168 (22.6)	0.002	9302 (25.3)	9203 (25.0)	0.007
VTE	285 (3.0)	290 (3.1)	0.006	285 (3.0)	293 (3.1)	0.006	1105 (3.0)	1120 (3.0)	<0.001
CHF	2026 (21.4)	2048 (21.6)	0.005	2144 (22.3)	2103 (21.9)	0.010	8924 (24.3)	8874 (24.2)	0.002
GERD	2536 (26.8)	2566 (27.1)	0.007	2638 (27.5)	2605 (27.1)	0.009	9905 (27.0)	9924 (27.0)	<0.001
Kidney failure	1442 (15.2)	1427 (15.1)	0.003	1455 (15.2)	1460 (15.2)	<0.001	6026 (16.4)	5923 (16.1)	0.008
Osteoporosis	689 (7.3)	709 (7.5)	0.008	689 (7.2)	734 (7.6)	0.015	3036 (8.3)	3015 (8.2)	0.004
Neurological disorders	530 (5.6)	521 (5.5)	0.004	555 (5.8)	525 (5.5)	0.013	2294 (6.2)	2268 (6.2)	<.001
Malignant neoplasm	1241 (13.1)	1222 (12.9)	0.006	1258 (13.1)	1257 (13.1)	<0.001	5609 (15.3)	5581 (15.2)	0.003
Metastatic solid tumor	273 (2.9)	264 (2.8)	0.006	262 (2.7)	262 (2.7)	<0.001	1155 (3.1)	1156 (3.1)	<0.001

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Table 1. Baseline Characteristics of Patients Initiating Umeclidinium-Vilanterol, Glycopyrrolate-Formoterol, or Tiotropium-Olodaterol in Propensity Score Matched Cohorts (continued)

Characteristic, No. (%)[a]	Cohort 1 Umeclidinium-vilanterol vs glycopyrrolate-formoterol (2017-2025; n = 9479 matched pairs)[b]			Cohort 2 Tiotropium-olodaterol vs glycopyrrolate-formoterol (2017-2025; n = 9598 matched pairs)[b]			Cohort 3 Umeclidinium-vilanterol vs tiotropium-olodaterol (2016-2025; n = 36 740 matched pairs)[b]		
	Umeclidinium-vilanterol	Glycopyrrolate-formoterol	Absolute standardized difference	Tiotropium-olodaterol	Glycopyrrolate-formoterol	Absolute standardized difference	Umeclidinium-vilanterol	Tiotropium--olodaterol	Absolute standardized difference
Anxiety disorders	2155 (22.7)	2187 (23.1)	0.010	2199 (22.9)	2201 (22.9)	<0.001	8076 (22.0)	8019 (21.8)	0.005
Depression	1950 (20.6)	1963 (20.7)	0.002	1999 (20.8)	1972 (20.5)	0.007	7198 (19.6)	7142 (19.4)	0.005
Stroke	33 (0.3)	23 (0.2)	0.020	22 (0.2)	23 (0.2)	<0.001	128 (0.3)	129 (0.4)	0.017
Lung cancer	610 (6.4)	591 (6.2)	0.008	621 (6.5)	601 (6.3)	0.008	2846 (7.7)	2819 (7.7)	<0.001
Health care utilization									
Emergency department visits									
0	6132 (64.7)	6209 (65.5)	0.017	6328 (65.9)	6306 (65.7)	0.004	23 842 (64.9)	23 837 (64.9)	<0.001
1	2016 (21.3)	1964 (20.7)	0.015	1940 (20.2)	1984 (20.7)	0.012	7745 (21.1)	7725 (21.0)	0.002
≥2	1331 (14.0)	1306 (13.8)	0.006	1330 (13.9)	1308 (13.6)	0.009	5153 (14.0)	5178 (14.1)	0.003
Hospitalizations									
0	7437 (78.5)	7488 (79.0)	0.012	7610 (79.3)	7584 (79.0)	0.007	28 183 (76.7)	28 205 (76.8)	0.002
1	1514 (16.0)	1489 (15.7)	0.008	1478 (15.4)	1502 (15.6)	0.006	6161 (16.8)	6137 (16.7)	0.003
≥2	528 (5.6)	502 (5.3)	0.013	510 (5.3)	512 (5.3)	<0.001	2396 (6.5)	2398 (6.5)	<0.001
Readmission (≤90 d)	493 (5.2)	471 (5.0)	0.001	461 (4.8)	478 (5.0)	0.009	2228 (6.1)	2227 (6.1)	<0.001
Office visits									
0	169 (1.8)	182 (1.9)	0.007	191 (2.0)	184 (1.9)	0.007	952 (2.6)	956 (2.6)	<0.001
1-5	4208 (44.4)	4179 (44.1)	0.006	4243 (44.2)	4217 (43.9)	0.006	15 769 (42.9)	15 822 (43.1)	0.004
6-10	3337 (35.2)	3345 (35.3)	0.002	3378 (35.2)	3398 (35.4)	0.004	13 101 (35.7)	13 092 (35.6)	0.002
≥11	1765 (18.6)	1773 (18.7)	0.003	1786 (18.6)	1799 (18.7)	0.003	6918 (18.8)	6870 (18.7)	0.003
Pulmonary office visit	1509 (15.9)	1436 (15.1)	0.022	1510 (15.7)	1469 (15.3)	0.011	5828 (15.9)	5853 (15.9)	<0.001
Unique prescriptions, mean (SD)	12.62 (6.37)	12.50 (6.33)	0.019	12.53 (6.18)	12.47 (6.28)	0.010	12.38 (6.12)	12.37 (6.05)	0.002
Basic or comprehensive metabolic blood chemistry test	6991 (73.8)	6953 (73.4)	0.009	7062 (73.6)	7054 (73.5)	0.002	28 063 (76.4)	27 941 (76.1)	0.007
ECG	4493 (47.4)	4493 (47.4)	<0.001	4558 (47.5)	4576 (47.7)	0.004	18 077 (49.2)	18 123 (49.3)	0.002
Echocardiogram	2239 (23.6)	2216 (23.4)	0.005	2247 (23.4)	2258 (23.5)	0.002	9449 (25.7)	9497 (25.8)	0.002
CT scan	3475 (36.7)	3462 (36.5)	0.004	3504 (36.5)	3508 (36.5)	<0.001	14 531 (39.6)	14 583 (39.7)	0.002
Bronchoscopy or biopsy	228 (2.4)	237 (2.5)	0.006	232 (2.4)	239 (2.5)	0.006	979 (2.7)	958 (2.6)	0.006
Mammography	1008 (10.6)	1034 (10.9)	0.010	1032 (10.8)	1049 (10.9)	0.003	3844 (10.5)	3850 (10.5)	<0.001
Colonoscopy and biopsy	23 (0.2)	32 (0.3)	0.020	32 (0.3)	31 (0.3)	<0.001	106 (0.3)	106 (0.3)	<0.001
Bone mineral density scan	436 (4.6)	449 (4.7)	0.005	479 (5.0)	465 (4.8)	0.009	1865 (5.1)	1903 (5.2)	0.005
Influenza vaccine	2999 (31.6)	3012 (31.8)	0.004	3113 (32.4)	3069 (32.0)	0.009	11 569 (31.5)	11 616 (31.6)	0.002

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Table 1. Baseline Characteristics of Patients Initiating Umeclidinium-Vilanterol, Glycopyrrolate-Formoterol, or Tiotropium-Olodaterol in Propensity Score Matched Cohorts (continued)

Characteristic, No. (%)[a]	Cohort 1			Cohort 2			Cohort 3		
	Umeclidinium-vilanterol vs glycopyrrolate-formoterol (2017-2025; n = 9479 matched pairs)[b]			Tiotropium-olodaterol vs glycopyrrolate-formoterol (2017-2025; n = 9598 matched pairs)[b]			Umeclidinium-vilanterol vs tiotropium-olodaterol (2016-2025; n = 36 740 matched pairs)[b]		
	Umeclidinium-vilanterol	Glycopyrrolate-formoterol	Absolute standardized difference	Tiotropium-olodaterol	Glycopyrrolate-formoterol	Absolute standardized difference	Umeclidinium-vilanterol	Tiotropium--olodaterol	Absolute standardized difference
Nonpulmonary medications									
Statins	5433 (57.3)	5335 (56.3)	0.020	5460 (56.9)	5461 (56.9)	<0.001	21 935 (59.7)	21 921 (59.7)	<0.001
β-Blockers	3756 (39.6)	3771 (39.8)	0.004	3852 (40.1)	3860 (40.2)	0.002	15 459 (42.1)	15 390 (41.9)	0.004
ACE inhibitors	2425 (25.6)	2409 (25.4)	0.005	2474 (25.8)	2450 (25.5)	0.007	9015 (24.5)	9000 (24.5)	<0.001
ARBs	2283 (24.1)	2217 (23.4)	0.016	2276 (23.7)	2260 (23.5)	0.005	8879 (24.2)	8906 (24.2)	<0.001
Calcium channel blockers	2781 (29.3)	2694 (28.4)	0.020	2777 (28.9)	2735 (28.5)	0.009	10 746 (29.2)	10 727 (29.2)	<0.001
Thiazide diuretics	1905 (20.1)	1888 (19.9)	0.005	1944 (20.3)	1913 (19.9)	0.010	6975 (19.0)	6939 (18.9)	0.003
Loop diuretics	2173 (22.9)	2176 (23.0)	0.002	2217 (23.1)	2224 (23.2)	0.002	9092 (24.7)	9065 (24.7)	<0.001
PPIs	3499 (36.9)	3406 (35.9)	0.021	3535 (36.8)	3478 (36.2)	0.012	13 453 (36.6)	13 356 (36.4)	0.004
H_2 receptor blockers	678 (7.2)	689 (7.3)	0.004	750 (7.8)	703 (7.3)	0.019	2802 (7.6)	2790 (7.6)	<0.001
Metformin	1466 (15.5)	1459 (15.4)	0.003	1474 (15.4)	1460 (15.2)	0.006	5183 (14.1)	5259 (14.3)	0.006
Sulfonylurea	536 (5.7)	516 (5.4)	0.013	522 (5.4)	525 (5.5)	0.004	2121 (5.8)	2114 (5.8)	<0.001
SGLT2	258 (2.7)	254 (2.7)	<0.001	259 (2.7)	262 (2.7)	<0.001	1353 (3.7)	1349 (3.7)	<0.001
DPP-4	300 (3.2)	287 (3.0)	0.012	264 (2.8)	285 (3.0)	0.012	994 (2.7)	1033 (2.8)	0.006
GLP-1 RAs	293 (3.1)	327 (3.4)	0.017	301 (3.1)	324 (3.4)	0.017	1238 (3.4)	1260 (3.4)	<0.001
Benzodiazepines	1955 (20.6)	1941 (20.5)	0.002	1957 (20.4)	1964 (20.5)	0.002	6728 (18.3)	6771 (18.4)	0.003
SSRIs and SNRIs	2926 (30.9)	2923 (30.8)	0.002	2920 (30.4)	2962 (30.9)	0.011	11 290 (30.7)	11 347 (30.9)	0.004
Socioeconomic covariates[b,e]									
Average pharmaceutical copay, mean (SD), $	20.79 (30.47)	21.34 (30.53)	0.018	21.10 (32.75)	21.44 (31.43)	0.011	21.54 (34.59)	21.45 (32.41)	0.003
Copay for pharmacy cost, mean (SD), $	302.19 (524.68)	306.46 (478.91)	0.009	305.33 (515.24)	306.80 (477.04)	0.003	304.60 (502.69)	304.04 (485.61)	0.001
Ratio of unique brand to generic drugs, mean (SD)	0.22 (0.31)	0.22 (0.30)	<0.001	0.22 (0.30)	0.22 (0.30)	<0.001	0.21 (0.29)	0.21 (0.29)	<0.001

Abbreviations: ACE, angiotensin-converting enzyme inhibitor; ARB, angiotensin-receptor blocker; BiPAP, bilevel positive airway pressure; CAD, coronary artery disease; CBC, complete blood count; CHD, coronary heart disease; CHF, congestive heart failure; COPD, chronic obstructive pulmonary disease; CPAP, continuous positive airway pressure; CT, computed tomography; DPP-4, dipeptidyl peptidase-4; ECG, electrocardiogram; GERD, gastroesophageal reflux disease; GLP-1 RA, glucagon-like peptide-1 receptor agonist; H_2, histamine 2; ICS, inhaled corticosteroid; IHD, ischemic heart disease; LABA, long-acting β_2-agonist; LAMA, long-acting muscarinic antagonist; PPI, proton pump inhibitor; PVD, peripheral vascular disease; SABA, short-acting β_2-agonist; SAMA, short-acting muscarinic antagonist; SGLT2, sodium–glucose cotransporter 2; SNRI, serotonin–norepinephrine reuptake inhibitor; SSRI, selective serotonin reuptake inhibitor; VTE, venous thromboembolism.

[a] Values are reported as numbers (percentages), unless otherwise indicated.

[b] In cohort 1, data were missing on sex for 3 patients, region for 2 patients, copays for 161 patients, and brand-to-generic ratios for 261 patients; in cohort 2, data were missing on sex for 3 patients, region for 3 patients, copays for 167 patients, and brand-to-generic ratios for 273 patients; in cohort 3, data were missing on sex for 7 patients, region for 38 patients, copays for 667 patients, and brand-to-generic ratios for 1084 patients. Percentages may not sum to 100 due to rounding.

[c] Data are presented together due to low counts in 2025. Years of initiation were adjusted for individually in a propensity score model.

[d] This category includes patients with at least 1 inpatient or outpatient claim coded for asthma using a look-back window that includes all available data (not just in prior 6-month baseline period).

[e] All socioeconomic covariates exclude day 0 from the assessment period and therefore exclude out-of-pocket payments for index prescriptions.

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Table 2. Effectiveness and Safety Outcome Effect Estimates in Propensity Score Matched Cohorts of New Combined LAMA-LABA Users

Outcome	Referent events per 1 person-year	Comparator events per 1 person-year	Hazard ratio (95% CI)
Umeclidinium-vilanterol (comparator) vs glycopyrrolate-formoterol (reference)			
Moderate or severe COPD exacerbation	0.83	0.68	0.86 (0.81-0.91)
Moderate COPD exacerbation	0.76	0.63	0.86 (0.81-0.91)
Severe COPD exacerbation	0.07	0.06	0.85 (0.71-1.02)
Pneumonia hospitalization	0.10	0.09	1.00 (0.86-1.16)
MACE	0.08	0.08	0.97 (0.83-1.14)
Urinary tract infection	0.02	0.02	0.92 (0.68-1.23)
Tiotropium-olodaterol (comparator) vs glycopyrrolate-formoterol (reference)			
Moderate or severe COPD exacerbation	0.81	0.75	0.94 (0.89-1.00)
Moderate COPD exacerbation	0.75	0.69	0.95 (0.89-1.00)
Severe COPD exacerbation	0.07	0.07	0.99 (0.83-1.18)
Pneumonia hospitalization	0.10	0.10	1.06 (0.91-1.22)
MACE	0.08	0.08	0.95 (0.80-1.10)
Urinary tract infection	0.02	0.02	0.94 (0.70-1.26)
Umeclidinium-vilanterol (comparator) vs tiotropium-olodaterol (reference)			
Moderate or severe COPD exacerbation	0.70	0.67	0.97 (0.94-0.99)
Moderate COPD exacerbation	0.65	0.62	0.97 (0.94-1.00)
Severe COPD exacerbation	0.06	0.06	0.91 (0.83-1.00)
Pneumonia hospitalization	0.10	0.10	1.06 (0.98-1.13)
MACE	0.09	0.09	1.02 (0.95-1.10)
Urinary tract infection	0.02	0.02	0.88 (0.76-1.02)

Abbreviations: COPD, chronic obstructive pulmonary disease; LABA, long-acting β_2-agonists; LAMA long-acting muscarinic antagonist; MACE, major adverse cardiovascular events (death, acute myocardial infarction, stroke).

These findings are consistent with a prior network meta-analysis that found umeclidinium-vilanterol was associated with a reduced risk of COPD exacerbations compared with both glycopyrrolate-formoterol and tiotropium-olodaterol (with similar pneumonia risk).[26] The indirect comparisons in that study raise the risk of bias due to heterogeneity in the populations, outcome definitions, and follow-up between the included studies. Our findings are also consistent with a small randomized trial[22] and a longitudinal database study[27] showing superiority of umeclidinium-vilanterol over tiotropium-olodaterol, although the primary end point of the former study was trough forced expiratory volume in the first second of expiration (FEV_1). The latter did not use propensity score matching, relied on prescribed rather than dispensed inhalers, and excluded patients receiving baseline ICS therapy, which may not be reflective of clinical practice.

The current study, to our knowledge, is the largest real-world study to date testing for potential intraclass differences among LAMA-LABAs. A number of hypotheses are available to explain the slight superiority of once-daily umeclidinium-vilanterol dry powder inhalers. Less frequent dosing regimens tend to be associated with improved adherence compared with regimens that require more frequent dosing.[45] In addition, randomized trials have found a patient-reported preference for umeclidinium-vilanterol dry powder inhalers due to their easy-to-use platforms.[46,47] The tiotropium-olodaterol soft mist inhaler and glycopyrrolate-formoterol metered-dosed inhaler, by contrast, require more steps for use, manual actuation, and/or hand-breath coordination, which could result in higher error rates that negatively impact drug delivery.[48-51] Whether it is because of less frequent dosing or lower error rates (or both), umeclidinium-vilanterol dry powder inhalers could have outperformed alternatives if patients initiating the regimen were more likely to (correctly) take their prescribed medications than those receiving metered-dosed inhaler or soft mist inhaler formulations. Although the different molecules under investigation could also be responsible for the observed findings, other studies have shown improved outcomes among patients with COPD receiving once-daily dry powder inhalers (in the case of ICS-LAMA-LABAs).[52,53]

In addition to a small potential clinical benefit, once-daily umeclidinium-vilanterol dry powder inhalers also have a substantial environmental benefit. Studies have documented significantly larger carbon footprints of metered-dosed inhalers due to hydrofluoroalkane propellants, which are potent greenhouse gases.[33,34,54,55] Although some patients may prefer or require metered-dosed inhalers, particularly frail older adults with poor inspiratory force, our study suggests that health systems may be able to promote umeclidinium-vilanterol dry powder inhalers as preferred therapy for most patients with COPD who are newly initiating LAMA-LABAs to advance environmental stewardship goals without undermining clinical outcomes. Studies

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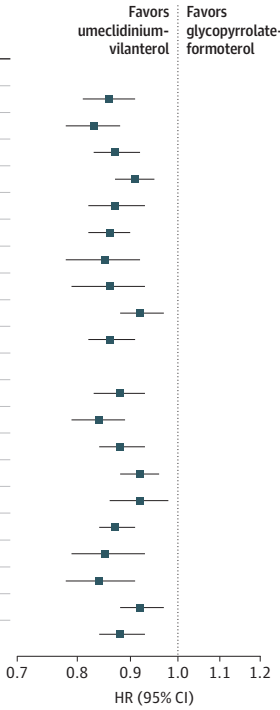
Figure 1. Sensitivity Analyses on Claims-Based Definitions for First Moderate or Severe Chronic Obstructive Pulmonary Disease Exacerbation

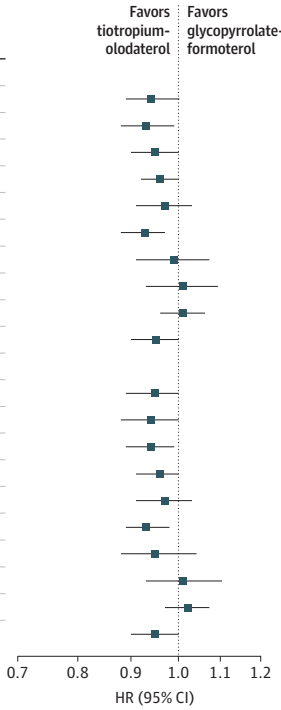
A Umeclidinium-vilanterol vs glycopyrrolate-formoterol

Analysis	HR (95% CI)
1:1 PS-matched cohort	
60-d Grace period[a]	0.86 (0.81-0.91)
30-d Grace period	0.83 (0.78-0.88)
90-d Grace period	0.87 (0.83-0.92)
As-started	0.91 (0.87-0.95)
Limit follow-up to 90 d	0.87 (0.82-0.93)
PS fine stratification	0.86 (0.82-0.90)
Require visit for moderate exacerbation	0.85 (0.78-0.92)
Require oral steroid + antibiotic for moderate exacerbation	0.86 (0.79-0.93)
Permit oral steroid or antibiotic for moderate exacerbation	0.92 (0.88-0.97)
Permit exacerbation code in any position for severe exacerbation	0.86 (0.82-0.91)
High-dimensional PS-matched cohort	
60-d Grace period	0.88 (0.83-0.93)
30-d Grace period	0.84 (0.79-0.89)
90-d Grace period	0.88 (0.84-0.93)
As-started	0.92 (0.88-0.96)
Limit follow-up to 90 d	0.92 (0.86-0.98)
High-demensional PS fine stratification	0.87 (0.84-0.91)
Require visit for moderate exacerbation	0.85 (0.79-0.93)
Require oral steroid + antibiotic for moderate exacerbation	0.84 (0.78-0.91)
Permit oral steroid or antibiotic for moderate exacerbation	0.92 (0.88-0.97)
Permit exacerbation code in any position for severe exacerbation	0.88 (0.84-0.93)



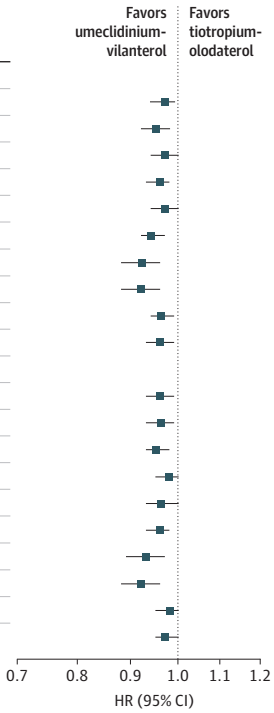
B Tiotropium-olodaterol vs glycopyrrolate-formoterol

HR (95% CI)
0.94 (0.89-1.00)
0.93 (0.88-0.99)
0.95 (0.90-1.00)
0.96 (0.92-1.00)
0.97 (0.91-1.03)
0.93 (0.88-0.97)
0.99 (0.91-1.07)
1.01 (0.93-1.09)
1.01 (0.96-1.06)
0.95 (0.90-1.00)
0.95 (0.89-1.00)
0.94 (0.88-1.00)
0.94 (0.89-0.99)
0.96 (0.91-1.00)
0.97 (0.91-1.03)
0.93 (0.89-0.98)
0.95 (0.88-1.04)
1.01 (0.93-1.10)
1.02 (0.97-1.07)
0.95 (0.90-1.00)



C Umeclidinium-vilanterol vs tiotropium-olodaterol

Analysis	HR (95% CI)
1:1 PS-matched cohort	
60-d Grace period[a]	0.97 (0.94-0.99)
30-d Grace period	0.95 (0.92-0.98)
90-d Grace period	0.97 (0.94-1.00)
As-started	0.96 (0.93-0.98)
Limit follow-up to 90 d	0.97 (0.94-1.00)
PS fine stratification	0.94 (0.92-0.97)
Require visit for moderate exacerbation	0.92 (0.88-0.96)
Require oral steroid + antibiotic for moderate exacerbation	0.92 (0.88-0.96)
Permit oral steroid or antibiotic for moderate exacerbation	0.96 (0.94-0.99)
Permit exacerbation code in any position for severe exacerbation	0.96 (0.93-0.99)
High-dimensional PS-matched cohort	
60-d Grace period	0.96 (0.93-0.99)
30-d Grace period	0.96 (0.93-0.99)
90-d Grace period	0.95 (0.93-0.98)
As-started	0.98 (0.95-1.00)
Limit follow-up to 90 d	0.96 (0.93-1.00)
High-demensional PS fine stratification	0.96 (0.93-0.98)
Require visit for moderate exacerbation	0.93 (0.89-0.97)
Require oral steroid + antibiotic for moderate exacerbation	0.92 (0.88-0.96)
Permit oral steroid or antibiotic for moderate exacerbation	0.98 (0.95-1.00)
Permit exacerbation code in any position for severe exacerbation	0.97 (0.95-1.00)



[a]Primary analysis results are presented in Table 2.

HR indicates hazard ratio; PS, propensity score.

are also needed to assess switching from one LAMA-LABA therapy to another because recent research has raised concerns about those who switch from ICS-LABA metered-dosed inhalers to dry powder inhalers.[56]

Limitations

This study has several limitations. First, we used data from patients enrolled in large commercial health insurance and Medicare Advantage plans, so the results may not be gener-

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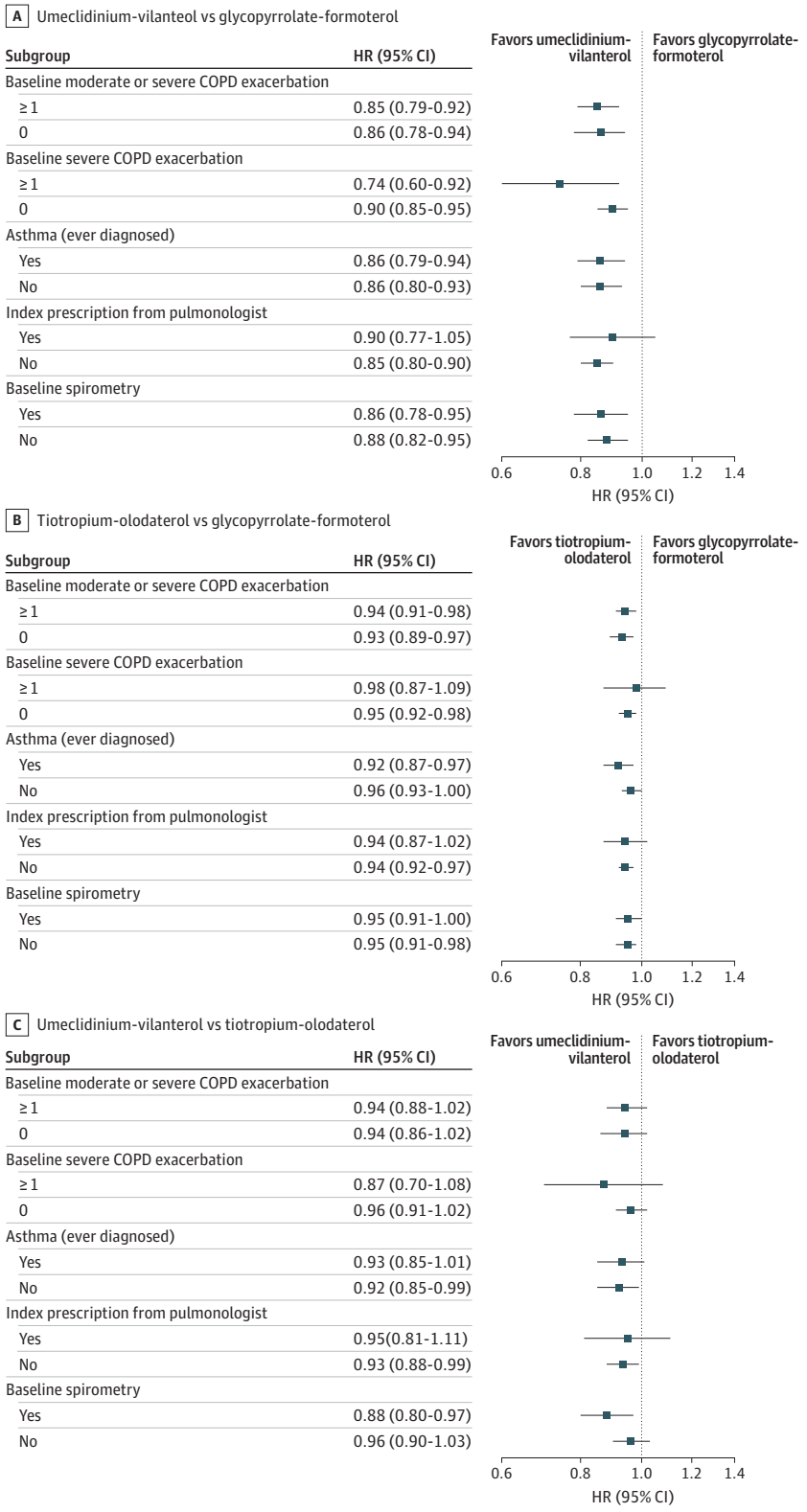
Figure 2. Subgroup Analyses for First Moderate or Severe Chronic Obstructive Pulmonary Disease Exacerbation



COPD indicates chronic obstructive pulmonary disease.

alizable to patients covered by other types of insurance. Second, we only assessed 3 LAMA-LABA inhalers; future analyses should expand to other less commonly used products. Third, a sizable percentage of patients were censored due to stopping the index exposure, and follow-up time was therefore relatively short. Although the as-started and

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90-day follow-up analyses showed similar estimates to the primary analyses, additional studies may assess time-varying and long-term effects of initiating these inhalers to validate sustained differences. Fourth, the comparison of tiotropium-olodaterol vs glycopyrrolate-formoterol and several subgroup analyses across all 3 cohorts were underpowered with wide confidence intervals. Fifth, the claims-based definitions of COPD and COPD exacerbations are subject to misclassification. We relied on prior validation studies and conducted a range of sensitivity analyses defining outcomes in different ways, which yielded similar results to the primary analysis; in addition, any misclassification was likely nondifferential and would tend to bias results toward the null. Sixth, patients may have filled prescriptions for medications that they did not take; we could only observe dispensing, not medication administration.

Finally, we adjusted for several covariates related to patients' lung disease and overall health, but we could not adjust for measures related to socioeconomic status, race, ethnicity, plan type, environmental exposures, or disease markers such as FEV_1 or emphysematous changes on computed tomographic scan. As a result, there could be residual confounding from unmeasured factors that may have predisposed clinicians to favor one inhaler over another. The choice of LAMA-LABA is often dictated by formulary design rather than provider or patient preference and, reassuringly, our sensitivity analysis using high-dimensional propensity score matching (which adjusts for proxies of unmeasured covariates) yielded nearly identical results as the primary analysis. Future studies, including randomized trials comparing LAMA-LABAs, could help further address questions of unmeasured confounding.

Conclusions

This study found that once-daily umeclidinium-vilanterol dry powder inhalers were associated with lower risk of moderate or severe COPD exacerbations compared with twice-daily glycopyrrolate-formoterol metered-dosed inhalers and once-daily tiotropium-olodaterol soft mist inhalers in patients with COPD treated in routine clinical practice. Patients, prescribers, payers, and health systems seeking to reduce greenhouse gas emissions may favor the once-daily umeclidinium-vilanterol dry powder inhaler over other LAMA-LABA inhalers for new users without compromising clinical outcomes.

ARTICLE INFORMATION

Accepted for Publication: December 10, 2025.

Published Online: February 23, 2026. doi:10.1001/jamainternmed.2025.8087

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Author Contributions: Drs Portela and Feldman had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.
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Acquisition, analysis, or interpretation of data: All authors.
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Obtained funding: Feldman.
Administrative, technical, or material support: Wang.
Supervision: Wang, Suissa, Feldman.

Conflict of Interest Disclosures: Dr Wang reported receiving grants from the National Institutes of Health (NIH) during the conduct of the study; personal fees from Cytel Inc, the ICA Group, and MITRE, a federally funded research and development center for the Centers for Medicare & Medicaid Services outside the submitted work. Dr Suissa reported serving on an advisory board for Atara Advisory, Novartis, and Panalgo and speaker fees from CSL Behring, and Covispharma outside the submitted work. Dr Feldman reported serving as a consultant for Alosa Health and as an expert witness in litigation against inhaler manufacturers outside the submitted work. No other disclosures were reported.

Funding/Support: This work was funded by grant K08HL163246 from the National Heart, Lung, and Blood Institute (Dr Feldman).

Role of the Funder/Sponsor: The National Heart, Lung, and Blood Institute had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Sharing Statement: See Supplement 2.

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